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# Simplified Circular Economy Indicator for Mining Operations: A Case in the Chilean Mining Industry

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## Abstract

The circular economy is becoming increasingly important in mining operations as the industry faces pressure to become more sustainable and responsible. This article proposes a circularity indicator to solve the problem of comparing mining operations over time. This indicator considers total production, CO<sub>2</sub> emissions, water consumption, and total energy used in production. The model responds to the industry's need to find indicators that are more adjusted to the data and less dependent on opinions. Results indicate that this indicator is relatively easy to implement and is demonstrated by comparing three mining plants over three years, evidencing the possibility of direct comparisons, and establishing the temporal progression of each mining plant.

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## 1. Introduction

Nowadays, one of the most concerning problems is the impact that human activity generates on the environment. This is because the reserves of resources that generate products are running out, and the population and its demand for goods are rapidly increasing. To explain the impact of human activity in a planetary context, it must be considered the predominance of linear production systems within the global economy, which have proven to be inefficient when managing natural non-renewable resources.

The linear production model is characterized by the conceptual phrase: “Extract, Produce and Discard”. For almost 300 years, industrial economies, in general, did not consider the impact generated by productive activity on the envi-

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ronment. In traditional economic theory, these impacts are assumed as one additional variable for the price fluctuations in the market, without considering costs in the strict sense of the definition.

Mineral resources are essential to allow the development of economic activities and the functioning of our society, as we know it. Stopping extracting resources and producing goods is neither a reasonable nor an ethical alternative. In recent years there has been a trend for companies to consider the Social Responsibility of belonging to a human fabric that not only examines people as economic entities, but as complex, unique beings in full exercise of their rights.

The so-called Linear Economy, associated with the industrial revolution, focused its attention on the massive generation of final products, trying to supply the needs for new goods. There was a perception that the environment could absorb all the changes generated by the industrial activity without considering the environmental effects. Today, because of the increase in the quantity and quality of communications, society has a clearer view about the production limitations regarding the limits and environmental impacts.

There is a growing interest on how to measure circular economy [1]. The predominant approaches include subjective studies (based on people's opinions), surveys and, in general, tools that collect series of data that are apparently unrelated, and usually are not accurately measurable.

The Circulytics tool from the Ellen MacArthur Foundation [2], whose database is built from a survey carried out within the productive units, aiming to obtain the circularity indicator. This approach avoids the possibility of comparing the functioning of different units and following the evolution of the company over time. If we examine the tool's usage graph in detail, we see that it is complex to apply, difficult to read, and has many indicators that are often overlapping.

Environmental impact and circularity implementation drivers and barriers are especially important for the mining industry. [3, 4, 5, 6, 7]

The annual reports in this sector includes numerical indicators regarding magnitudes that are relevant, allowing the quantification of environmental impacts. These indicators include water consumption, Carbon Dioxide  $CO_2$  emissions, fuel and electricity consumption, and total production (copper or other metals). Additional data could be incorporated in later stages of the analysis, such as productivity, total area used, tons of processed metal, total waste volumes, smelting gases, sulfuric acid consumption, among others.

This paper aims to propose a circularity indicator (overall efficiency) for mining operations considering available information for consumptions, emissions, and production within a set of mining plants. This paper is organized as follows: Section 2 includes a description and the main considerations of the proposed circularity indicator. Section 3 shows the results for the indicator calculations. Finally, section 4 presents the main conclusions.

## 2. Methodology

The proposed indicator aims to measure the level of simple circularity, based on available data from the mining plants of one the most important mining companies in Chile. For the correct implementation of this indicator, an in-depth study is not required to understand the circular economy and its principles, also analyzing the different measurement methods, circularity and identities linked to the environmental subjects and optimization of resources. This differs from the Circulytics approach and other similar tools and the idea for this indicator came from some aspects of the extensive work done by [8, 9], regarding circular economy in mining and metallurgy.

The indicator aims to allow a mining company to adopt different approaches in terms of its actual consumption and emissions, reaching a transformation towards a successful and efficient circular economy and allowing comparing the results with other companies in the sector. The proposed solution is based on measurable physical magnitudes that are informed and that are available in the annual management reports of mining companies. The study is focused on a Copper mining company in Chile, selected considering the authors' familiarity with the field, the relative importance of the sector within the Chilean economy and the evidently linear economic vision of mining, which extracts non-renewable resources.

This model is conceptually based on [10], which associates  $CO_2$  emissions with economic growth, population, and energy consumption. The proposed model (Equation 1) provides a global circularity indicator associated with copper production, fresh water consumed,  $CO_2$  emissions and electrical energy dissipated in the production process.

$$\text{Overall Efficiency} = (\text{Water consumption} * \text{Energy consumption} * (\text{CO}_2 \text{ Emissions}) / \text{Production})^3 \quad (1)$$

With Equation (1) is possible to obtain a global efficiency of each plant per year, where the 3 indicators that were compiled are considered (Water - Energy -  $CO_2$  Emissions), managing to identify with this, how efficiently they use their resources and emissions based on production made annually. The units of measurement of the variables are Megaliters of Water (MI), GigaJoules (GJ) of Energy consumed, Tons of equivalent  $CO_2$  produced and Tons of fine Copper produced, so the indicator has its own set of units that is:  $(MI \times GJ \times Ton CO_2) / (ton^3)$ . This metrological unit is an important difference with other subjective indicators of circularity and allows adding value to circular economy analyses.

This model does not consider productivity, hours worked or other numerical indicators. This is because the incorporation of new analysis variables can be considered indirectly by the variables that are incorporated. In a very particular way, this indicator does not consider the opinions of people about their own perception of circularity. This allows the release of opinions that may be far from the real behavior of the company since they are based on perceptions of people and not on data that comes from measurements of physical magnitudes.

Another relevant concept of this model is its simplicity. It is possible to obtain an approximation quickly and simply to one of the missing data given that the formula allows solving one of the unknowns in a trivial way and using indicators from similar companies or from previous periods to estimate the unmeasured variable. It also allows quantifying the impact on circularity by varying the flows measured prior to the variation and globally managing the management of the mining company.

### 3. Results

Calculations for the indicator, use the data published in the sustainability reports, considering three mining plants in Chile: Cerro Colorado, Spence and Escondida. Table 1 shows the data for the input variables of the indicator from 2018 to 2020.

Table 1. Input data for the Circularity Indicator

|                | Year | Water consumption<br>(MI) | Energy<br>consumption<br>(GJ) | $CO_2$ emissions<br>(ton) | Copper Production<br>(ton) |
|----------------|------|---------------------------|-------------------------------|---------------------------|----------------------------|
| Cerro Colorado | 2018 | 2,060                     | 3,168,698                     | 275,000                   | 66,279                     |
|                | 2019 | 3,867                     | 3,093,003                     | 258,000                   | 77,710                     |
|                | 2020 | 4,010                     | 2,976,032                     | 244,000                   | 68,899                     |
| Spence         | 2018 | 6,954                     | 4,443,329                     | 628,000                   | 176,377                    |
|                | 2019 | 6,750                     | 4,766,303                     | 665,000                   | 193,379                    |
|                | 2020 | 6,924                     | 4,358,710                     | 600,000                   | 146,630                    |
| Escondida      | 2018 | 102,171                   | 31,544,580                    | 4,197,000                 | 1,242,687                  |
|                | 2019 | 100,790                   | 31,067,057                    | 4,028,000                 | 1,156,875                  |
|                | 2020 | 100,392                   | 32,334,055                    | 4,282,000                 | 1,155,854                  |

Table 2 shows the results after using Equation (1). The main usefulness provided by this indicator is allowing the comparison between mining plants and tracking the evolution of the circularity, as showed in Figure 1.

Regarding the results of the model, Figure 1 present the values of the circularity indicator and show that it allows the comparison between different relative sizes of the mining plants for the same year and considering the evolution of the indicator over time.

Table 2. Results for the Circularity indicator

|                | 2018  | 2019  | 2020  |
|----------------|-------|-------|-------|
| Cerro Colorado | 6,165 | 6,576 | 8,903 |
| Spence         | 3,537 | 2,959 | 5,744 |
| Escondida      | 7,049 | 8,146 | 9,001 |

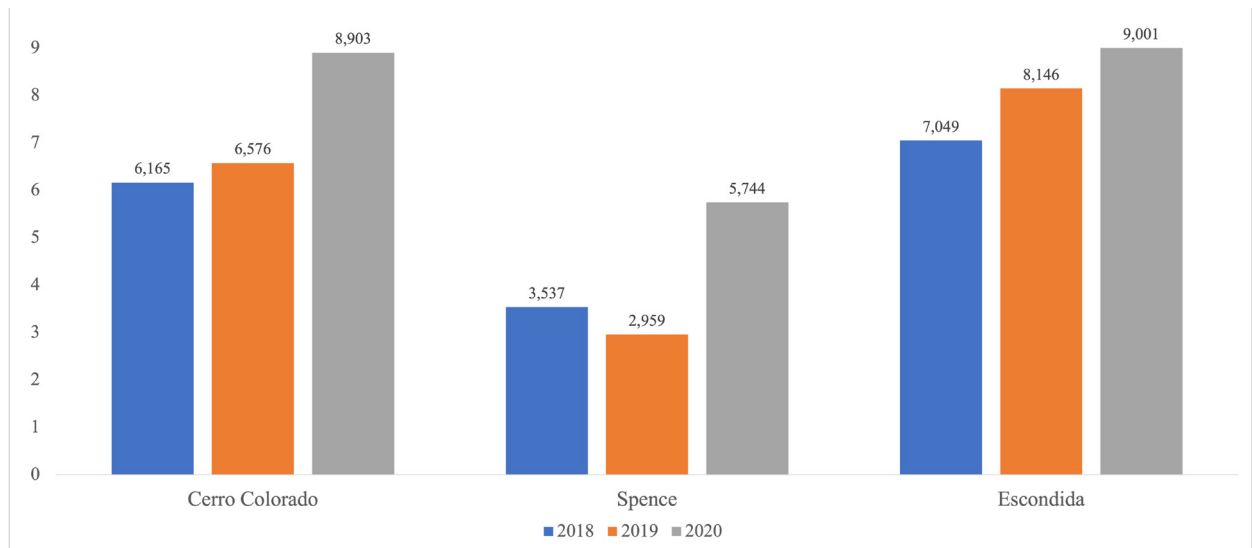


Fig. 1. Circularity indicator over time

#### 4. Conclusions

The paper proposed a numerical circularity indicator that allowed to obtain relationships regarding the global efficiency per year for some mining plants for one of the most important mining companies in Chile. Three factors (Water - Energy -  $CO_2$  Emission) were considered, helping to evaluate emissions and how efficiently the resources are being used, considering the total annual production.

It is possible to compare the performance of the three different mining plants and the temporal evolution of each of them over three years. Although the illustrative case of three mining plants and 2018-2020 period demonstrates the expected potential of the indicator, the analysis could be extended to more plants and consider recent years.

Finally, the circularity measurement may be incorporated in a simple and intuitive way to the key performance indicators (KPIs) of mining companies. The next challenge is including other variables without affecting the simplicity of the model and adding value to the analysis of dimensions currently not considered by the model. This integration allows decision-makers to monitor the performance and progress of mining plants simply and intuitively.

However, it is recognized that some additional variables and dimensions can influence the circularity of mining operations. The future direction of this research lies in expanding the model to incorporate these variables without compromising its simplicity. The analysis can cover aspects not currently considered, increasing the value of evaluating mining practices.

In conclusion, the numerical indicator of circularity presented in this article offers valuable information on the global efficiency of mining plants. Its integration into KPIs has excellent potential for mining companies looking to improve their sustainability performance. As this research serves as a foundation, future research may expand on it by broadening the scope of analysis and exploring new dimensions, leading to a more comprehensive understanding of minerals.

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## References

- [1] O. Marinina, N. Kirsanova, M. Nevskaya, Circular economy models in industry: Developing a conceptual framework, *Energies* 15 (2022) 9376. doi:10.3390/en15249376.
- [2] E. M. Foundation, *How to build a circular economy*, Ellen MacArthur Foundation (2022).  
URL <https://ellenmacarthurfoundation.org/>
- [3] S. Santoro, A. H. Avci, M. Aquino, L. Pugliese, S. Straface, E. Curcio, *Towards the global rise of zero liquid discharge for wastewater management: The mining industry case in chile*, *The Handbook of Environmental Chemistry* 118 (2021) 505–518. doi:10.1007/698\_2021\_785.  
URL [https://link.springer.com/10.1007/698\\_2021\\_785](https://link.springer.com/10.1007/698_2021_785)
- [4] R. A. Hodge, M. Ericsson, O. Löf, A. Löf, P. Semkowich, The global mining industry: corporate profile, complexity, and change, *Mineral Economics* 35 (2022) 587–606. doi:10.1007/s13563-022-00343-1.
- [5] Y. Li, H. Cen, T.-Y. Lin, Y.-N. Lin, Y.-H. Chiu, Sustainable coal mine and coal land development in china, *Resources Policy* 79 (2022) 103092. doi:10.1016/j.resourpol.2022.103092.
- [6] J. P. Vargas, A. P. M. Lagos, J. G. F. Torres, I. S. Kinney, S. P. Cortés, J. P. Hurtado, Achieving circularity through novel product-service systems in the mining industry: An opportunity for circularity, *Sustainability* 14 (2022) 3614. doi:10.3390/su14063614.
- [7] Éléonore Lèbre, G. Corder, A. Golev, The role of the mining industry in a circular economy: A framework for resource management at the mine site level, *Journal of Industrial Ecology* 21 (2017) 662–672. doi:10.1111/jiec.12596.
- [8] J. M. Huerta, R. C. Montecino, M. S. Medina, *Una aproximación termodinámica para la comprensión de la economía circular aplicada al ámbito minero - metalúrgico*, *Revista de Medio Ambiente y Minería* 6 (2021) 26–32.  
URL [http://www.scielo.org.bo/scielo.php?script=sci\\_arttext&pid=S2519-53522021000100003&lng=es&nrm=iso&tlng=es](http://www.scielo.org.bo/scielo.php?script=sci_arttext&pid=S2519-53522021000100003&lng=es&nrm=iso&tlng=es)
- [9] M. Sanchez, Use of copper slag for iron alloys fabrication. a circular economy approach for the chilean metallurgical industry, *The 11th Molten Slags, Fluxes and Salts [MOLTEN 2020]* (5 2020).
- [10] S. M. Eskander, J. Nitschke, Energy use and co2 emissions in the uk universities: An extended kaya identity analysis, *Journal of Cleaner Production* 309 (2021) 127199. doi:10.1016/j.jclepro.2021.127199.